

Decision Modelling (NPTEL Online Course 2021)

Assignment 1 (Decision Analysis – L1-L5)

Detailed Answers

- 1) _____ type of decision makers use a structured procedure of developing and analyzing decision alternatives before selecting the best choice. Whereas _____ type of decision makers use incomplete and imperfect information and tend to satisfice:
- i. **Normative, Descriptive**
 - ii. Descriptive, Normative
 - iii. Descriptive, Descriptive
 - iv. Normative, Normative
- 2) Under the decision making environment of uncertainty:
- i. Decision-making is straightforward as there is no ambiguity
 - ii. Decision-making is at risk because of ambiguities due to the known probabilities of outcomes
 - iii. **Decision-making is uncertain as the probabilities of outcomes are not known**
 - iv. None of these
- 3) Which of the following statements correctly describes the comparison between Expected value of perfect information (EVPI) and the minimum expected regret (MER):
- i. EVPI is always greater than MER
 - ii. EVPI is always less than MER
 - iii. EVPI is either greater than or less than MER
 - iv. **EVPI is always equal to MER**
- 4) In *Maximax* decision making criterion, the decision maker:
- i. **Compares the maximum payoffs for each alternatives and selects the one with maximum payoffs**
 - ii. Compares the minimum payoffs for each alternatives and selects the one with maximum payoffs
 - iii. Considers the best state of nature and selects the alternative with the maximum payoff in this state of nature
 - iv. Considers the worst state of nature and selects the alternative with the maximum payoff in this state of nature

5) In *Maximin* decision making criterion, the decision maker:

- i. Compares the maximum payoffs for each alternatives and selects the one with maximum payoffs
- ii. **Compares the minimum payoffs for each alternatives and selects the one with maximum payoffs**
- iii. Considers the best state of nature and selects the alternative with the maximum payoff in this state of nature
- iv. Considers the worst state of nature and selects the alternative with the maximum payoff in this state of nature

6) Hurwicz decision making criteria can be the same as that of Maximax criteria when degree of optimism value is:

- i. 0
- ii. 0.5
- iii. **1**
- iv. Cannot be said

7) For the upcoming cultivation season, a farmer has the option of cultivating one out of four variety of rice. The payoffs associated with the farming depends on the amount of rainfall in the monsoon. The estimated payoff matrix is given below:

		Annual profit in thousands (State of nature)			
		Heavy rain	moderate rain	light rain	draught
Cultivation options	Variety 1	-20	60	30	-5
	Variety 2	40	50	35	0
	Variety 3	-50	100	45	-10
	Variety 4	12	15	15	10

Choose the right decision based on *Maximin Criterion*.

- i. Variety 1
- ii. Variety 2
- iii. Variety 3
- iv. **Variety 4**

Answer to Q7

		Annual profit in thousands (State of nature)				Minimum payoff
		Heavy rain	moderate rain	light rain	draught	
Cultivation options Decision	Variety 1	-20	60	30	-5	-20
	Variety 2	40	50	35	0	0
	Variety 3	-50	100	45	-10	-10
	Variety 4	12	15	15	10	10

Maximum of minimum payoffs will be 10. So, the decision will be to cultivate **variety 4**.

- 8) A manufacturing company want to invest in purchasing new machinery. The following table indicates the payoffs associated with different investment decision alternatives.

Decision Alternatives	Annual profit in lakhs (State of nature)		
	Strong economy	Stable economy	Weak economy
High capacity	300	80	-100
Medium capacity	120	90	-40
Small capacity	40	30	20

Choose the right decision based on *Maximax Criterion*.

- i. **High capacity**
- ii. Medium capacity
- iii. Small capacity

Answer to Q8

Decision Options	Annual profit in lakhs (State of nature)			Maximum payoffs
	Strong economy	Stable economy	Weak economy	
High capacity	300	80	-100	300
Medium Capacity	120	90	-40	120
Small Capacity	40	30	20	40

Maximum of maximum payoffs will be 300.

Therefore, the decision will be to go for **high capacity**.

- 9) Using the table given in Question no 8, choose the right decision based on Hurwicz Criterion if the 'Degree of Optimism' is 0.6.

- i. **High capacity**
- ii. Medium capacity
- iii. Small capacity

Answer to Q9

	State of nature			Maximax Payoffs	Maximin Payoffs	Hurwicz Payoffs DOF =0.6
	Strong economy	Stable economy	Weak economy			
High capacity	300	80	-100	300	-100	$0.6*300+0.4*(-100) = \mathbf{140}$
Medium capacity	120	90	-40	120	-40	$0.6*120 + 0.4*(-40) = 56$
Small capacity	40	30	20	40	20	$0.6*40 + 0.4*(20) = 32$

Therefore, the decision will be **high capacity**

- 10) Using the table given in Question 8, choose the right decision based on Laplace Criterion.

- i. **High capacity**
- ii. Medium capacity
- iii. Small capacity

Answer to Q10

		Annual profit in lakhs (State of nature)			Calculation of payoffs
		Strong economy	Stable economy	Weak economy	
Investment options	High capacity	300	80	-100	$300*1/3 + 80*1/3 -100*1/3 = \mathbf{93.33}$
	Medium capacity	120	90	-40	$120*1/3 + 90*1/3 -40*1/3 = 56.67$
	Small capacity	40	30	20	$40*1/3 + 30*1/3 +20*1/3 = 30.00$

Under 'Laplace Criterion' all the states of nature are equally likely. As, there are three states of nature, the probability of each state of nature will be 1/3.

Maximum expected payoff is 93.33 for the decision **high capacity**.

11) For the payoff matrix given below, which decision will be taken based on *Expected Value Approach*?

		Annual profit in lakhs (State of nature)		
		Strong economy	Stable economy	Weak economy
Probability of Occurrence		0.6	0.3	0.1
Investment options	High capacity	300	80	-100
	Medium capacity	120	90	-40
	Small capacity	40	30	20

- i. **High capacity**
- ii. Medium capacity
- iii. Small capacity

Answer to Q11

		Annual profit in lakhs (State of nature)			Calculation of payoffs
		Strong economy	Stable economy	Weak economy	
Probability of Occurrence		0.6	0.3	0.1	
Investment options / Decision	High capacity	300	80	-100	$300 \times 0.6 + 80 \times 0.3 + (-100) \times 0.1$ = 194
	Medium capacity	120	90	-40	$120 \times 0.6 + 90 \times 0.3 + (-40) \times 0.1$ = 95
	Small capacity	40	30	20	$40 \times 0.6 + 30 \times 0.3 + 20 \times 0.1$ = 35

In 'Expected Value Approach', we multiply the payoff with probability of occurrence of state of nature for each decision alternatives and add them all.

The maximum expected payoff of 194 is obtained for the decision **High capacity**.

12) What is the probability of getting a sum 9 from two throws of a dice?

- i. 1/6
- ii. 1/8
- iii. **1/9**
- iv. 1/12

Answer to Q12

In two throws of a dice, total number of outcomes $n(S) = (6 \times 6) = 36$.

Let E = event of getting a sum 9 = $\{(3, 6), (4, 5), (5, 4), (6, 3)\}$.

$$\text{Then, } n(E) = 4; \quad \therefore P(E) = \frac{n(E)}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

13) If A and B are independent events, $P(A \cap B) =$

- i. $P(A)/P(B)$
- ii. $P(A)P(B)$**
- iii. $P(B)/P(A)$
- iv. $P(A)+P(B)$

14) A card is drawn from a pack of 52 cards. The probability of getting a queen of club or a king of heart is:

- i. $1/13$
- ii. $2/13$
- iii. $1/26$**
- iv. $1/52$

Answer to Q14

Here, $n(S) = 52$.

Let E = event of getting a queen of club or a king of heart.

$$\text{Then, } n(E) = 2. \quad \therefore P(E) = \frac{n(E)}{n(S)} = \frac{2}{52} = \frac{1}{26}$$

15) A bag contains 6 black and 8 white balls. One ball is drawn at random. What is the probability that the ball drawn is white?

- i. $3/4$
- ii. $4/7$**
- iii. $1/8$
- iv. $3/7$

Answer to Q15.

Let number of balls $n(S) = (6 + 8) = 14$. Number of white balls $n(E) = 8$.

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{8}{14} = \frac{4}{7}$$

16) A box contains three coins: two regular coins and one fake coin with both side marked head. If a coin is picked at random and tossed, what is the probability that the outcome is head?

i. $1/3$

ii. $2/3$

iii. $1/2$

iv. 1

Answer to Q16.

Let A be the event that regular coin is selected; B be the event the fake coin is selected

H be the event the outcome is head

Now $P(A) = 2/3$; $P(B) = 1/3$; $P(H|A) = 1/2$; $P(H|B) = 1$

Required probability = $P(H|A)*P(A) + P(H|B)*P(B) = (1/2)*(2/3) + 1*(1/3) = 2/3$

17) A fair die is rolled twice to obtain two numbers. Let X_1 = result of the first roll and X_2 = result of the second roll. If $X_1 + X_2 = 7$, what is the probability that $X_1 = 4$ or $X_2 = 4$?

i. $1/6$

ii. $2/3$

iii. $1/3$

iv. $1/36$

Answer to Q17.

Let A be the event that $X_1 = 4$ or $X_2 = 4$ and B be the event that $X_1 + X_2 = 7$.

We are interested in $P(A|B)$, so we can use $P(A|B) = P(A \cap B) / P(B)$

$A = \{(4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (1,4), (2,4), (3,4), (5,4), (6,4)\}$,

$B = \{(6,1), (5,2), (4,3), (3,4), (2,5), (1,6)\}$,

$A \cap B = \{(4,3), (3,4)\}$. $P(A|B) = P(A \cap B) / P(B) = (2/36) / (6/36)$

$= 1/3$

18) Two students are selected from a class to the quiz team. If one of the student selected is a girl, what is the probability that both the students selected are girls?

- i. $1/4$
- ii. $1/3$
- iii. $1/2$
- iv. $2/3$

Answer to Q18.

Sample space $S = \{(G, G), (G, B), (B, G), (B, B)\}$

Let A be the event that both students are girls, i.e., $A = \{(G, G)\}$.

B be the event that at least one of the students is a girl, i.e., $B = \{(G, G), (G, B), (B, G)\}$.

Since the outcomes are equally likely, we can write $P(A) = 1/4$, $P(B) = 3/4$.

Required probability, $P(A|B) = P(A \cap B) / P(B) = P(A) / P(B)$ (since $A \subset B$)

$$= (1/4) / (3/4) = 1/3.$$

19) Tickets numbered 1 to 20 are mixed up and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?

- i. $1/2$
- ii. $3/5$
- iii. $9/20$
- iv. $8/15$

Answer to Q19.

Here, $S = \{1, 2, 3, 4, \dots, 19, 20\}$.

Let E = event of getting a multiple of 3 or 5 = $\{3, 5, 6, 9, 10, 12, 15, 18, 20\}$.

$$P(E) = n(E) / n(S) = 9/20.$$

20) Two friends Jack and John appear in an interview for two vacancies in the same post. The probability of Jack's selection is $(1/7)$ and the probability of John's selection is $(1/5)$. What is the probability that only one of them is selected?

- i. **2/7**
- ii. $1/7$
- iii. $3/4$
- iv. $4/5$

Answer to Q20.

A = event that Jack is selected and B = event that John is selected

$$P(A) = 1/7; \quad P(B) = 1/5$$

$$\begin{aligned} \text{Required probability} &= P(A \text{ and not } B) + P(B \text{ and not } A) = (1/7) * (4/5) + (1/5) * (6/7) \\ &= 10/35 = \mathbf{2/7} \end{aligned}$$